Two Samples of Service Times

AP Statistics · Unit 4 · Topic 4.7 · B: 2027-01-25 · E: 2027-03-12 · TEACHER KEY

CED TETHER

- **Skill 1.D** — Identify an appropriate inference method for confidence intervals.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **Skill 3.D** — Construct a confidence interval, provided conditions for inference are met.
- **EU UNC-4** — An interval of values should be used to estimate parameters, in order to account for uncertainty.
- **LO UNC-4.V** — Identify an appropriate confidence interval procedure for a difference of two population means. [Skill 1.D]
- **EK UNC-4.V.1** — Consider a simple random sample from population 1 of size n_1, mean μ_1, and standard deviation σ_1 and a second simple random sample from population 2 of size n_2, mean μ_2, and standard deviation σ_2. If the distributions of populations 1 and 2 are normal or if both n_1 and n_2 are greater than 30, then the sampling distribution of the difference of means, $\bar{x}_1$ - $\bar{x}_2$, is also normal. The mean for the sampling distribution of $\bar{x}_1$ - $\bar{x}_2$ is μ_1 - μ_2. The standard deviation of $\bar{x}_1$ - $\bar{x}_2$ is $\sqrt{}(\sigma_1^2/n_1 + \sigma_2^2/n_2)$.
- **EK UNC-4.V.2** — The appropriate confidence interval procedure for one quantitative variable for two independent samples is a two-sample t-interval for a difference between population means.
- **LO UNC-4.W** — Verify the conditions to calculate confidence intervals for the difference of two population means. [Skill 4.C]
- **EK UNC-4.W.1** — In order to calculate confidence intervals to estimate a difference of population means, we must check for independence and that the sampling distribution is approximately normal:
- (a) To check for independence: (i) Data should be collected using two independent, random samples or a randomized experiment. (ii) When sampling without replacement, check that $n_1 \leq 10\%\ N_1$ and $n_2 \leq 10\%\ N_2$.
- (b) To check that the sampling distribution of ($\bar{x}_1$ - $\bar{x}_2$) should be approximately normal (shape): (i) If the observed distributions are skewed, both n_1 and n_2 should be greater than 30.
- **LO UNC-4.X** — Determine the margin of error for the difference of two population means. [Skill 3.D]
- **EK UNC-4.X.1** — For the difference of two sample means, the margin of error is the critical value (t*) times the standard error (SE) of the difference of two means.
- **EK UNC-4.X.2** — The standard error for the difference in two sample means with sample standard deviations s_1 and s_2 is $\sqrt{}(s_1^2/n_1 + s_2^2/n_2)$.
- **LO UNC-4.Y** — Calculate an appropriate confidence interval for a difference of two population means. [Skill 3.D]
- **EK UNC-4.Y.1** — The point estimate for the difference of two population means is the difference in sample means, $\bar{x}_1$ - $\bar{x}_2$.
- **EK UNC-4.Y.2** — For a difference of two population means where the population standard deviations are not known, the confidence interval is ($\bar{x}_1$ - $\bar{x}_2$) ± t* $\sqrt{}(s_1^2/n_1 + s_2^2/n_2)$ where ±t* are the critical values for the central C% of a t-distribution with appropriate degrees of freedom that can be found using technology.

Essential question: How does the sampling design determine whether a paired or independent-means confidence interval is trustworthy?

DOK-3 skill: 4.C · **FRQ pattern:** `design-selects-two-mean-interval`

DOK rationale: Part (c) links the original sampling design to competing standard errors and intervals, diagnoses response-based pairing, and traces how false precision changes the population conclusion.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3
Minutes: 45

Teacher does

- Board slide up at the bell. Ask what links existed before service times were observed.
- Begin the lesson video follow-along at minute 5.
- Return at minute 33. Require students to connect each standard error to a design.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to one procedure and identify a design detail still needed.
- Complete the follow-along, finish (a)–(c), revise, and turn in.

Questions to ask

- What link existed between a North record and a South record before either time was seen?
- Why can sorting the response values make the standard deviation of differences look small?
- Which standard error preserves the original sampling design?
- Which population can the random samples represent?

Adult role
Solo teacher. Trace design to standard error, interval width, interpretation, and scope.

[WARN] WATCH FOR

- Calling equal sample sizes a matched-pairs design.
- Accepting response-based sorted pairs as if they were planned pairs.
- Reading an interval containing 0 as proof that the means are equal.

SCORING PART (c) — E / P / I

Expected elements

- **selects-two-sample-procedure** (required) — Uses a two-sample t interval for separate SRSs
- **uses-competing-intervals** (required) — Contrasts $(-1.57, 5.97)$ with $(1.18, 3.22)$
- **diagnoses-response-pairing** (required) — Explains that sorting responses creates artificial association
- **traces-false-precision** (required) — Links $s_d = 3.0$ and $SE = 0.5$ to the changed conclusion
- **interprets-and-bounds-scope** (required) — Interprets the valid interval for the sampled monthly populations

E	Selects the design-based procedure, contrasts both intervals, explains false precision, and gives a scoped interpretation.
P	Selects the valid interval but incompletely explains the uncertainty change, conclusion, or scope.
I	Treats sorted ranks as planned pairs or uses $(1.18, 3.22)$ as design-based evidence.

[STAR] TODAY'S PROBLEM — annotated

Suppose two depots independently take SRSs of 36 weekday service-time records from monthly populations of 900 North and 1,200 South records. Technology gives $df = 69.31$ and $t^* = 1.995$ for a 95% two-sample interval.

Tariq: "I sorted both samples and paired equal ranks. The resulting differences have $\bar{d} = 2.2$ and $s_d = 3.0$, giving $(1.18, 3.22)$. Pairing makes the estimate more precise."

Mei: "The records were sampled separately, so I would use a two-sample interval. Sorting observed responses may change the uncertainty rather than reveal design-based pairs."

Service-time samples (min)

Depot	N	n	$\bar{x}$	s
North	900	36	42.3	8.4
South	1,200	36	40.1	7.6

FIRST TAKE (5 min)

Which interval procedure matches the original design? Cite any link that did or did not exist before times were observed.

Key: Not graded. Students should identify whether any pairing existed before the responses were sorted.

Rules callout "LET THE DESIGN CHOOSE THE INTERVAL (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Define $\mu_N - \mu_S$, calculate its point estimate, and name the design-based interval procedure.

Key: The parameter is the North-minus-South population mean service-time difference for that month's records. The estimate is $42.3 - 40.1 = 2.2$ minutes. Two independent SRSs require a two-sample t interval.

DOK 2 **(b)** Verify random, 10 percent, and shape conditions. Compute SE, margin of error, and the 95% interval.

Key: The independent SRSs satisfy $36 \leq 0.10(900) = 90$ and $36 \leq 0.10(1{,}200) = 120$. Both $n = 36 > 30$, supporting shape. $SE = \sqrt{8.4^2/36 + 7.6^2/36} = 1.88797$ and $ME = 1.995(1.88797) = 3.7665$. Thus the interval is $2.2 \pm 3.7665 = (-1.5665, 5.9665)$ minutes.

DOK 3 **(c)** Adjudicate the procedures using the separate SRSs, rank pairing, $s_d = 3.0$, and both intervals. Explain how response-based pairing changes uncertainty and the conclusion a reader might draw; then interpret the warranted interval and bound its population scope.

Key: Mei's procedure is warranted. No North record was linked to a South record before observing times. Sorting responses manufactures association; $s_d = 3.0$ yields the artificial paired $SE = 3/\sqrt{36} = 0.5$ and narrow interval $(1.18, 3.22)$, creating false precision and an apparently positive difference. The design-based interval is $(-1.57, 5.97)$. We are 95% confident the North-minus-South mean for the two depots' weekday records that month lies in that range. The independent SRSs support those finite monthly populations, not every transaction or a causal process effect.

EXIT

One thing the video changed about my first take: ______________________________